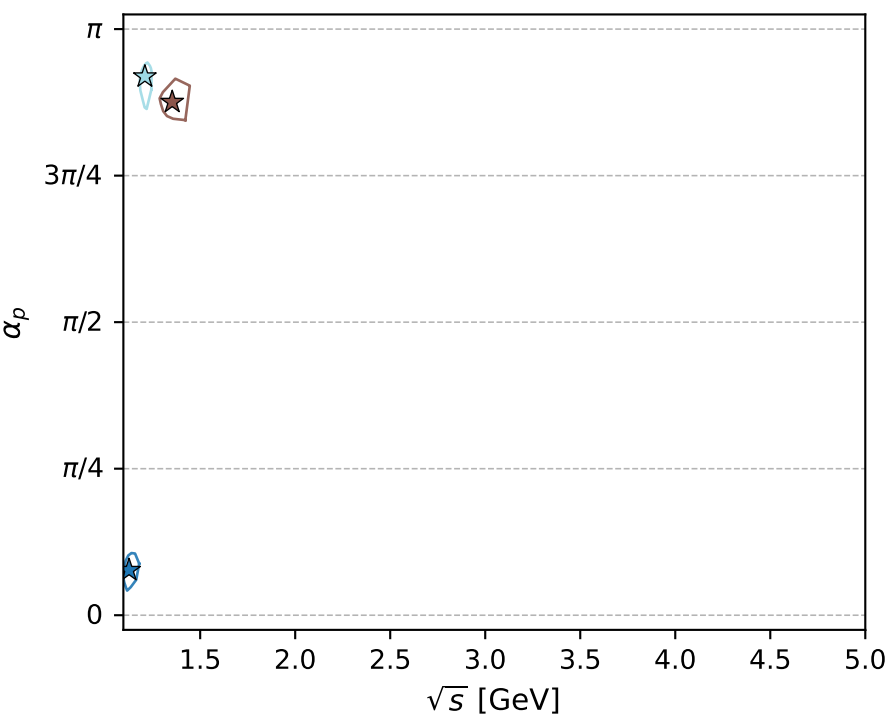
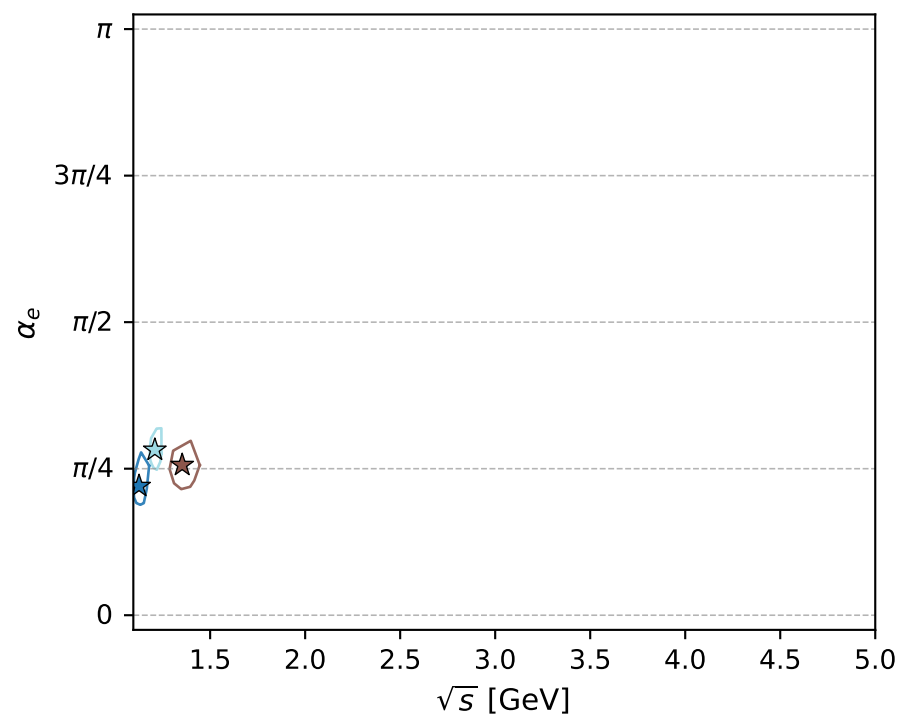
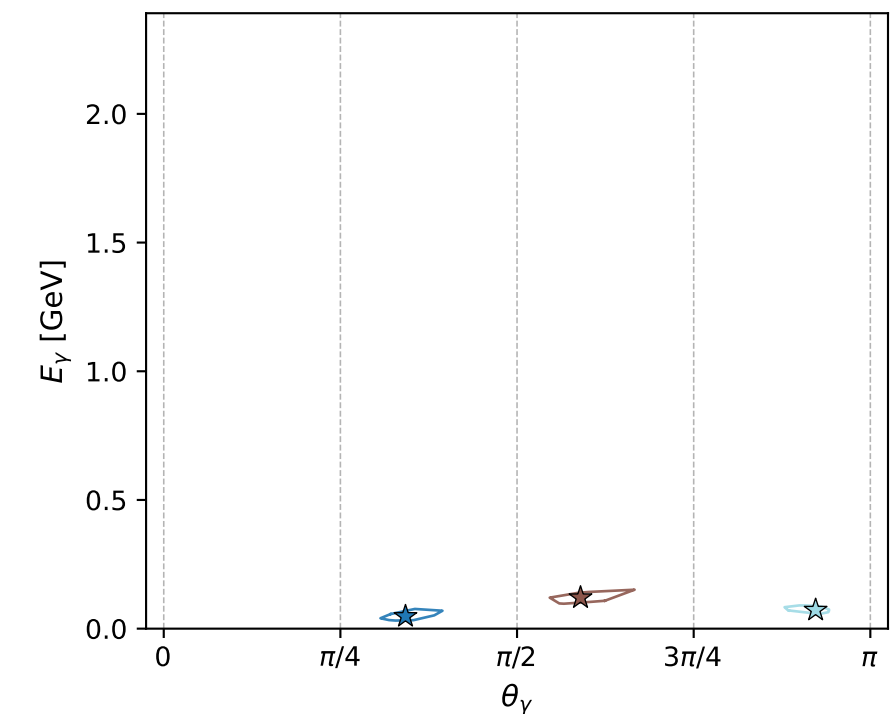
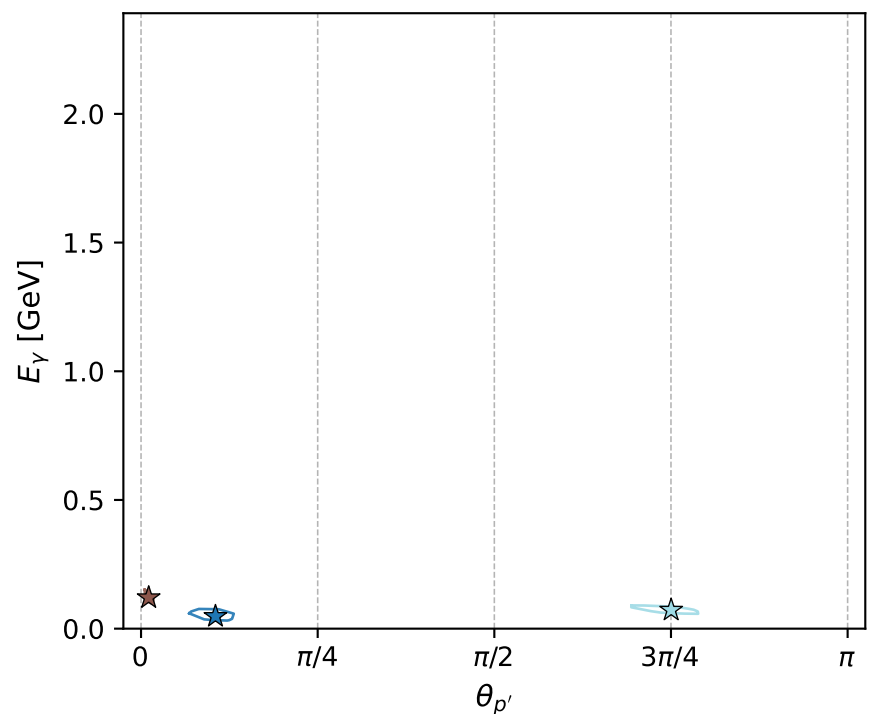
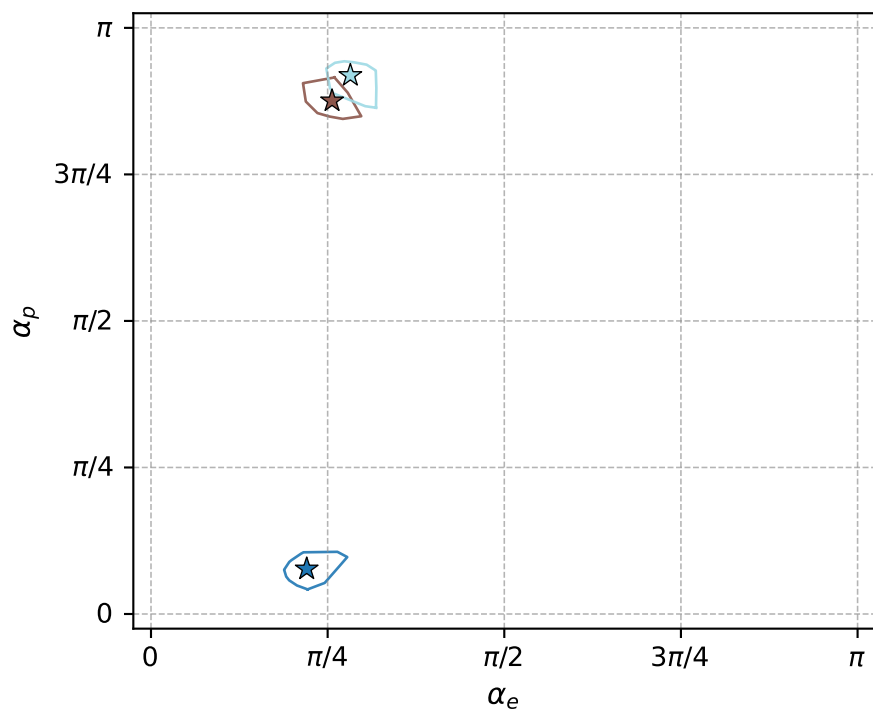
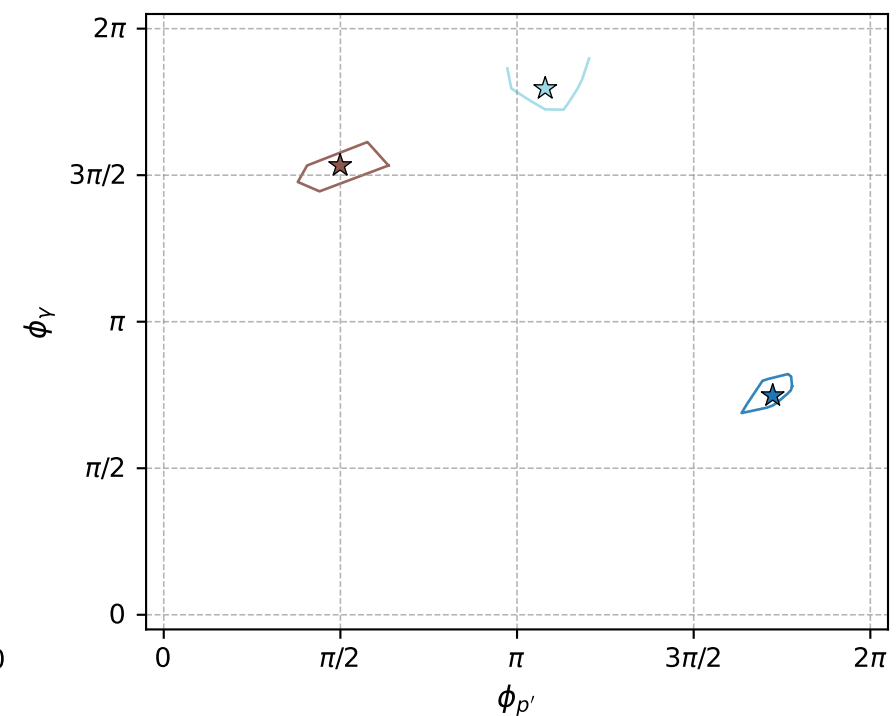
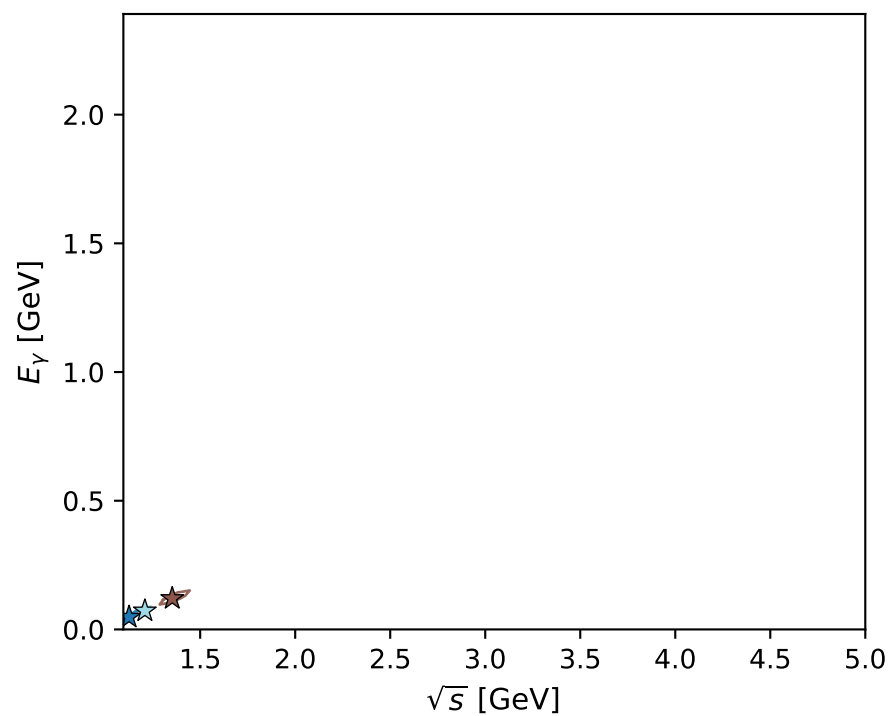
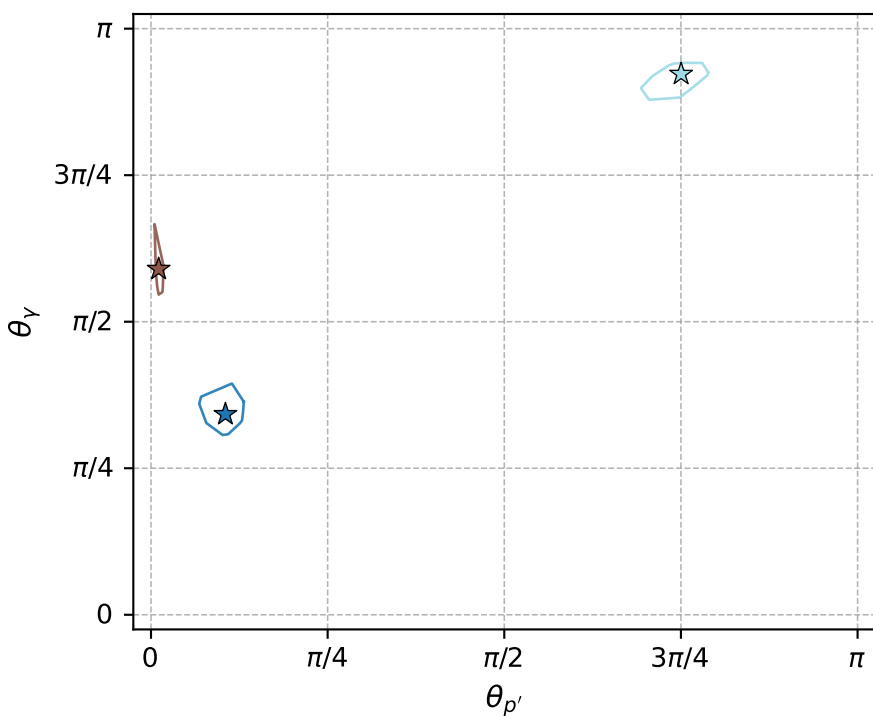


muon: polarization cluster P3; all local-minimum configurations and pairwise projections of their 8D contours



polarization cluster P3: $\alpha_e \sim \pi/4, \alpha_p \sim \pi$
 3 local minima in this polarization cluster
 contour delta=0.05
 configured 8D samples/minimum=6561
 D_W range=0.001511891 to 0.003975043
 global-minimum offsets=0.001508358 to 0.00397151

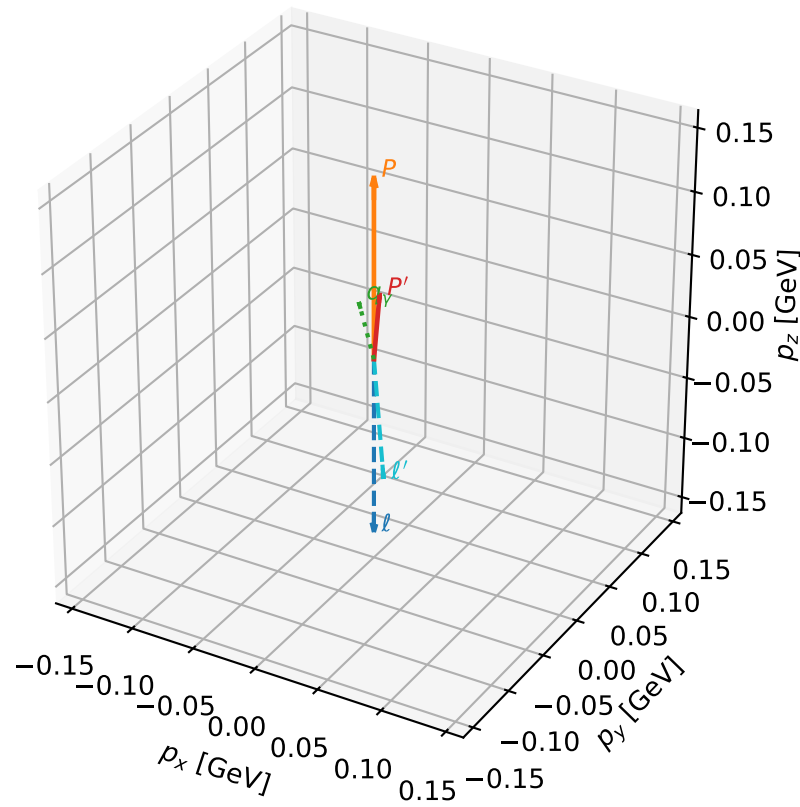
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_584 D_W=0.00151189
 region: polarization_cluster_03_local_minimum_584
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.69238412$ rad
 incoming lepton: $0.769726|+\rangle + 0.638374|-\rangle$
 proton mixing angle: $\alpha_p=0.24170928$ rad
 incoming proton: $0.97093|+\rangle + 0.239363|-\rangle$

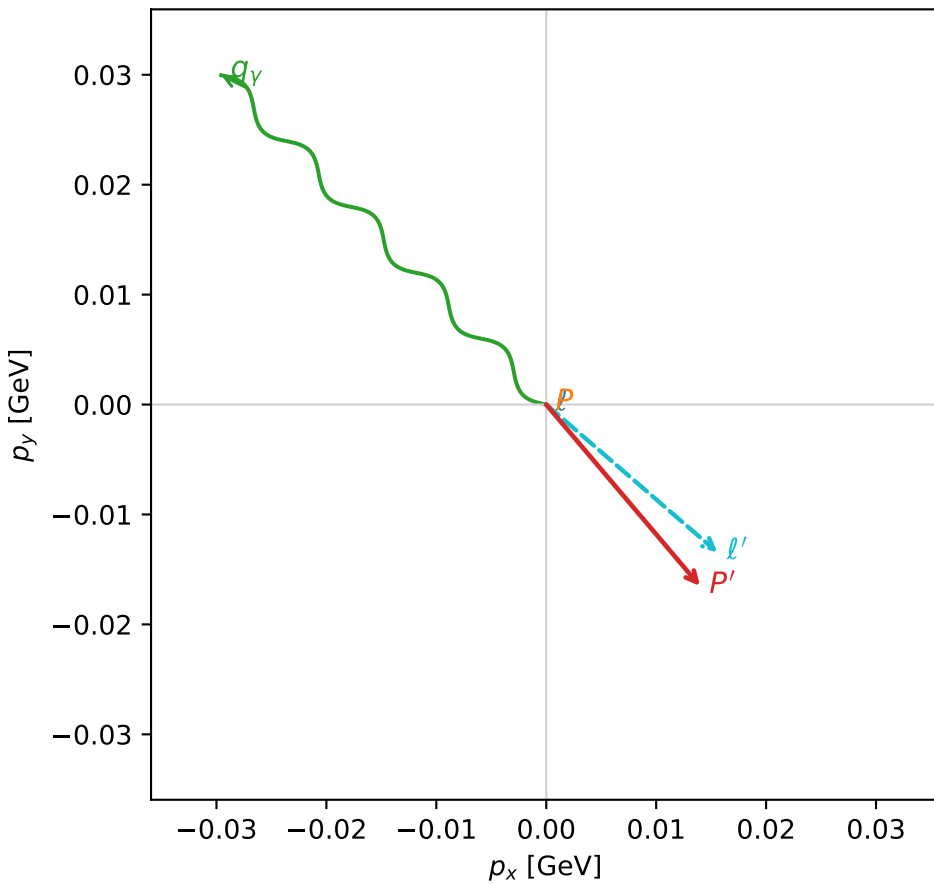
$s=1.26757$, $\sqrt{s}=1.12586$, $|\vec{P}|=0.142189$, $|\vec{P}'|=0.0666177$
 $\theta_{P'}=0.330885$, $\theta_V=1.07483$, $\phi_{P'}=5.41521$, $\phi_V=2.3496$
 $E_V=0.0478593$, $\theta_{\ell'}=2.90615$, $\phi_{\ell'}=5.57111$
 $Q^2=0.00204477$, $x_B=0.0224695$, $t=-0.00666888$, $W^2=0.968801$, $y=0.241666$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.1771$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.1422$
 P $E= 0.9487$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.1422$
 ℓ' $E= 0.1376$ $p_x= 0.0156$ $p_y= -0.0134$ $p_z= -0.0858$
 P' $E= 0.9404$ $p_x= 0.0140$ $p_y= -0.0165$ $p_z= 0.0630$
 q_V $E= 0.0479$ $p_x= -0.0296$ $p_y= 0.0300$ $p_z= 0.0228$

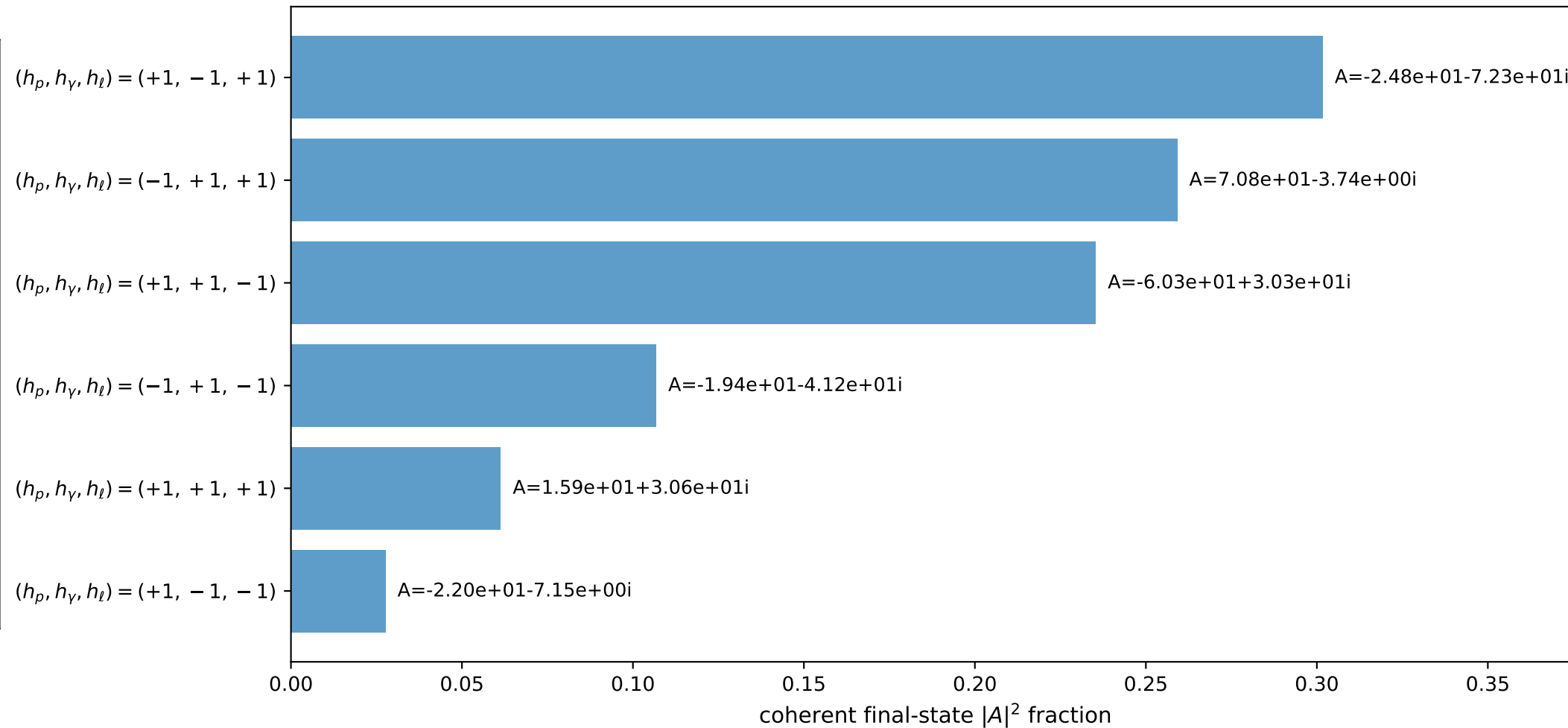
3D momenta



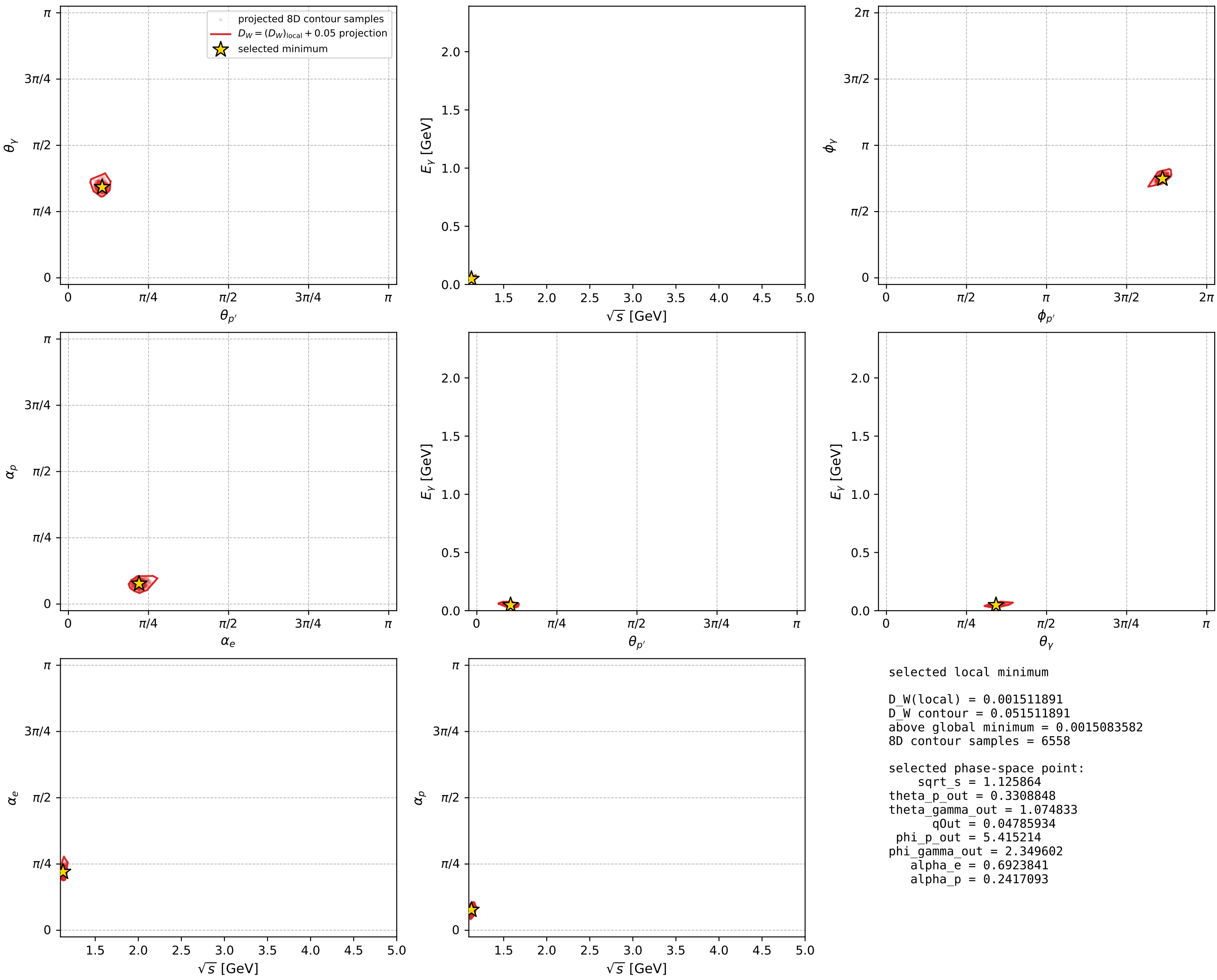
Transverse plane



Leading final-state amplitudes (N=6, retained=99.2%)



muon: polarization cluster P3, local minimum 584; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour

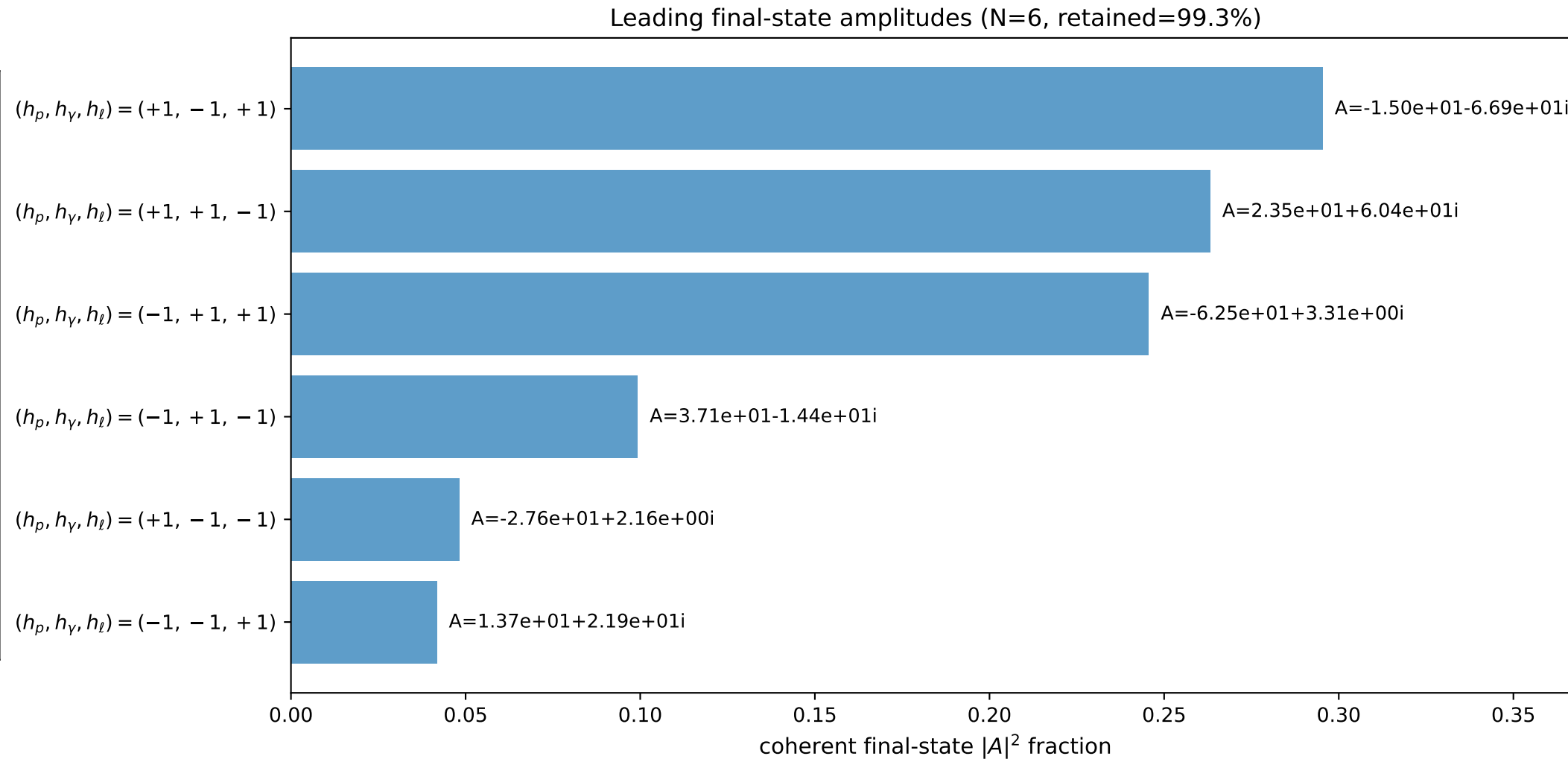
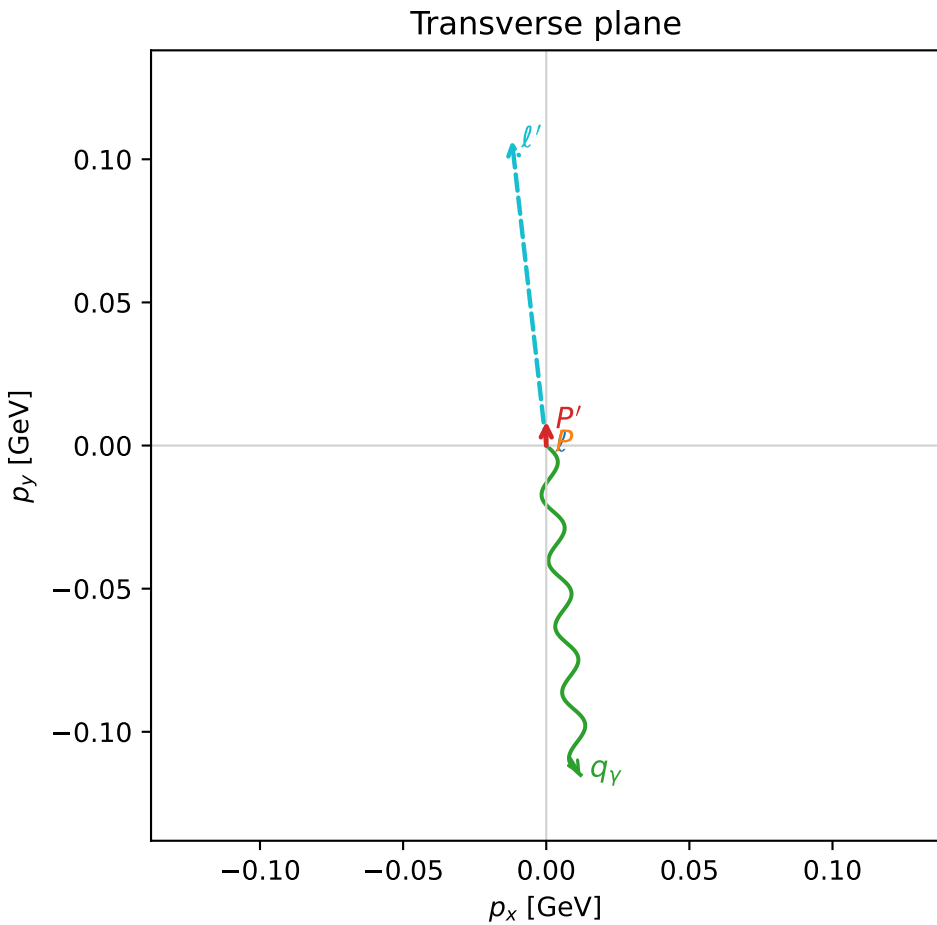
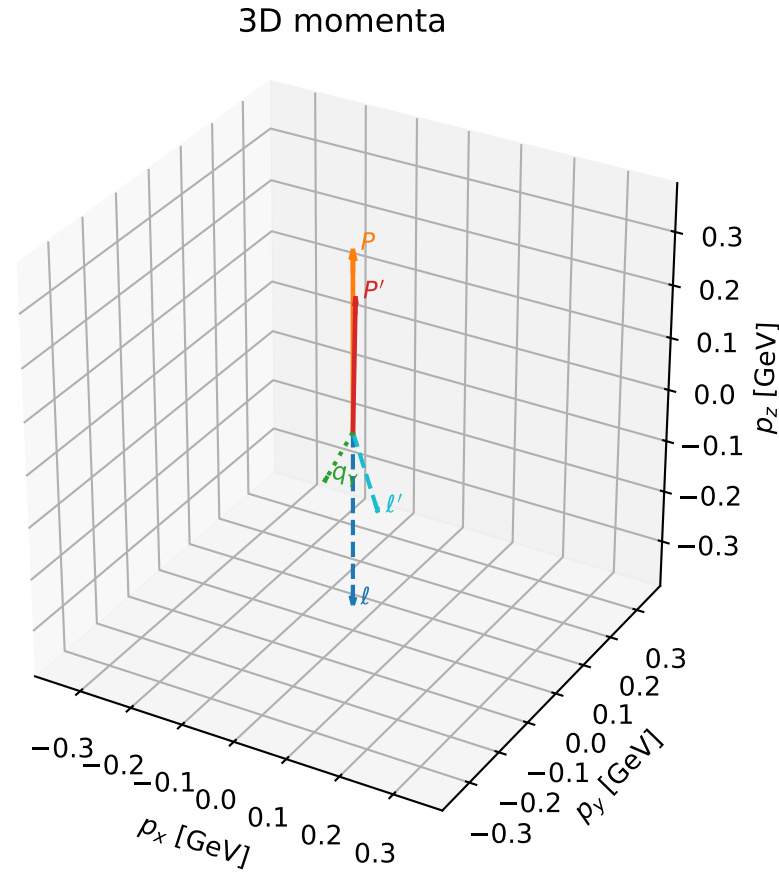


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

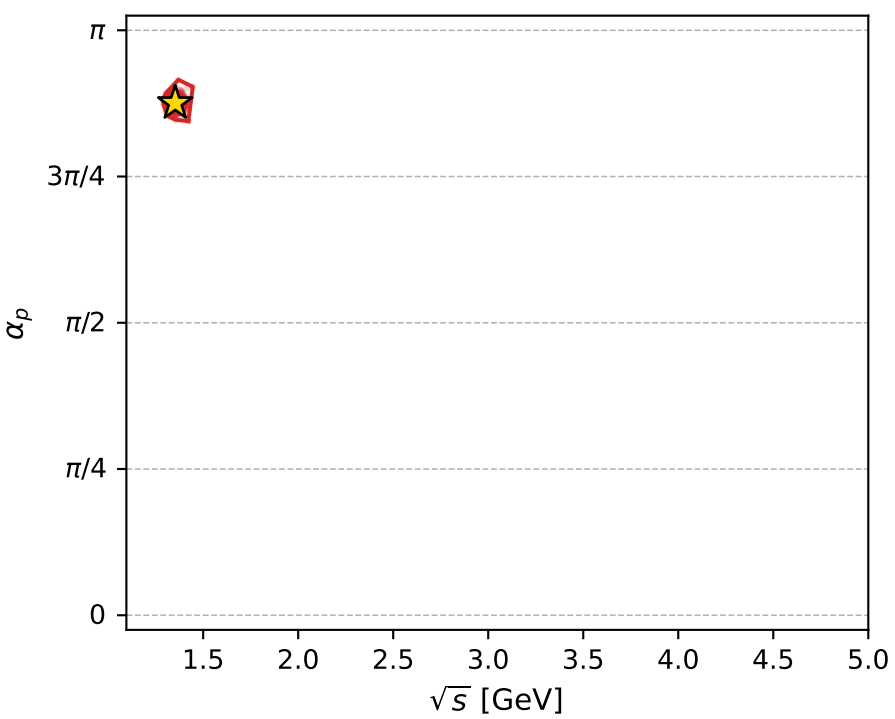
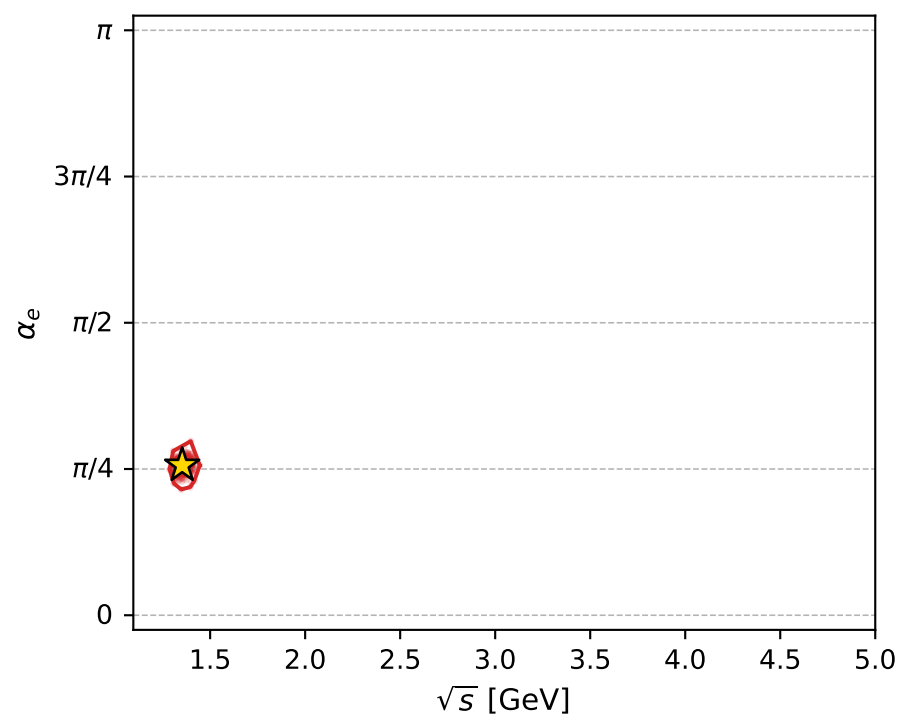
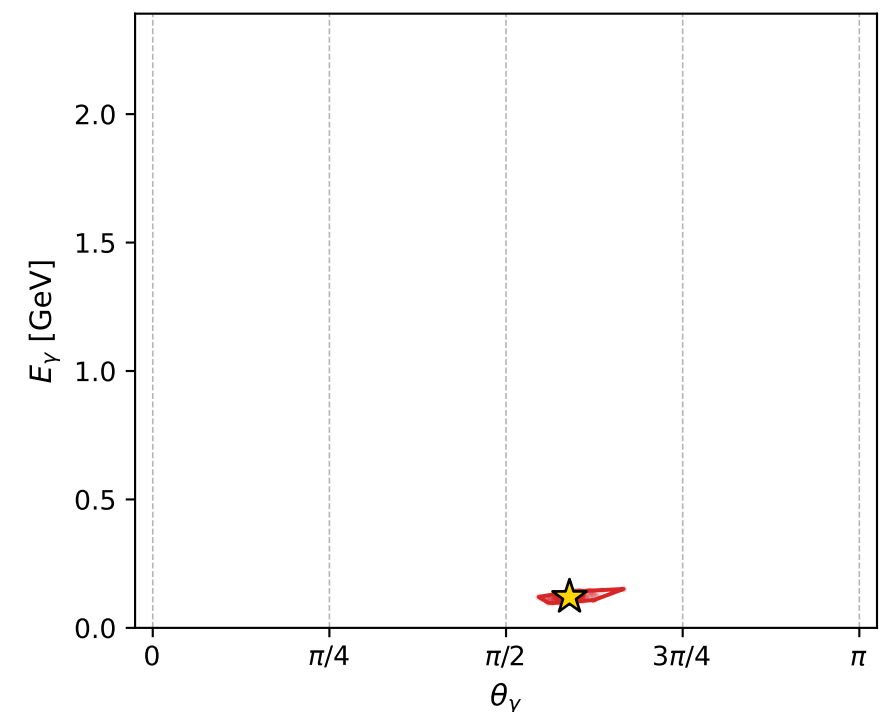
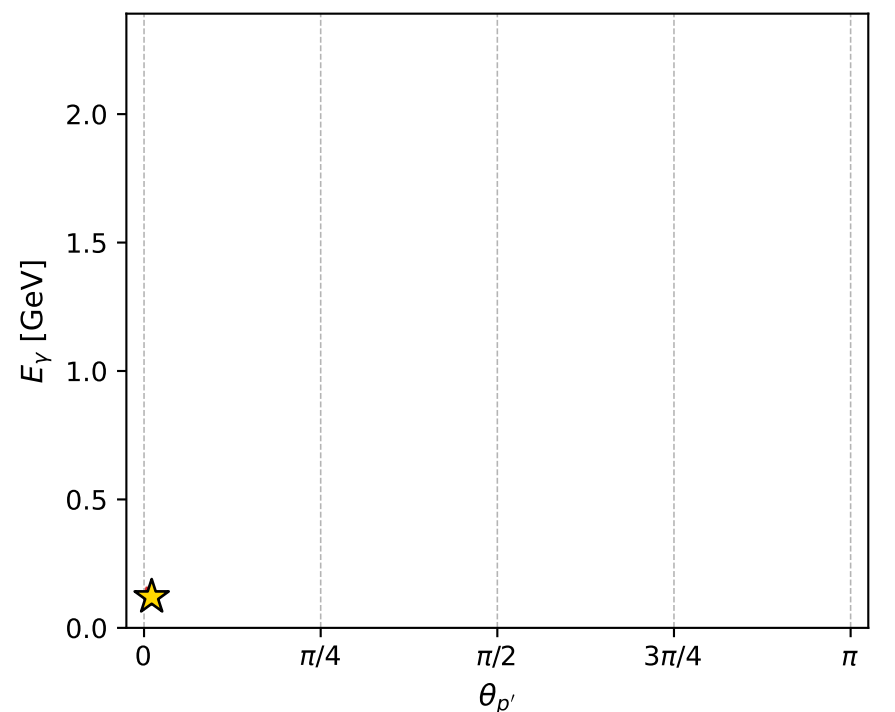
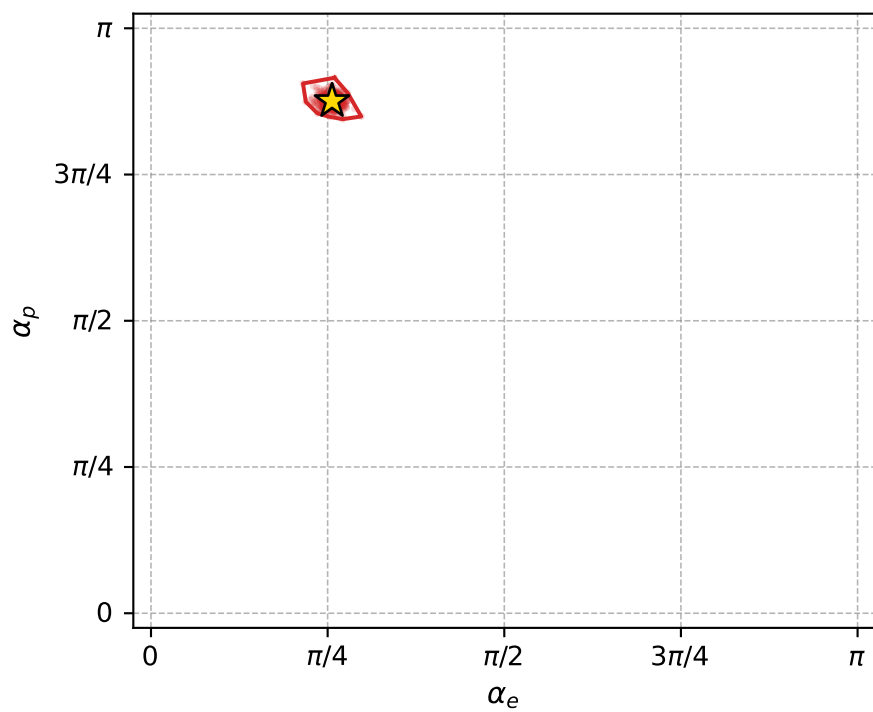
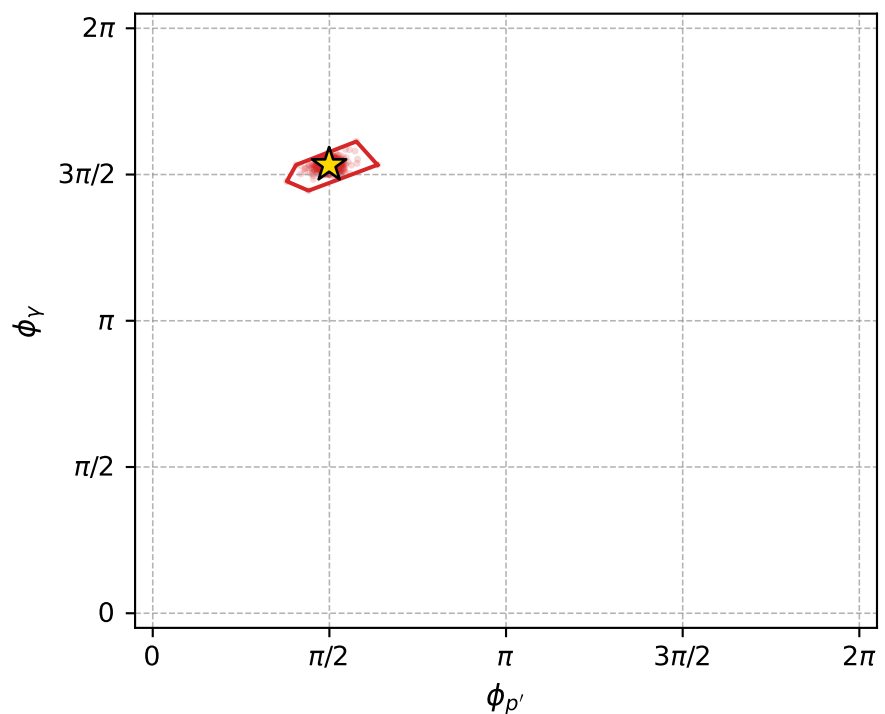
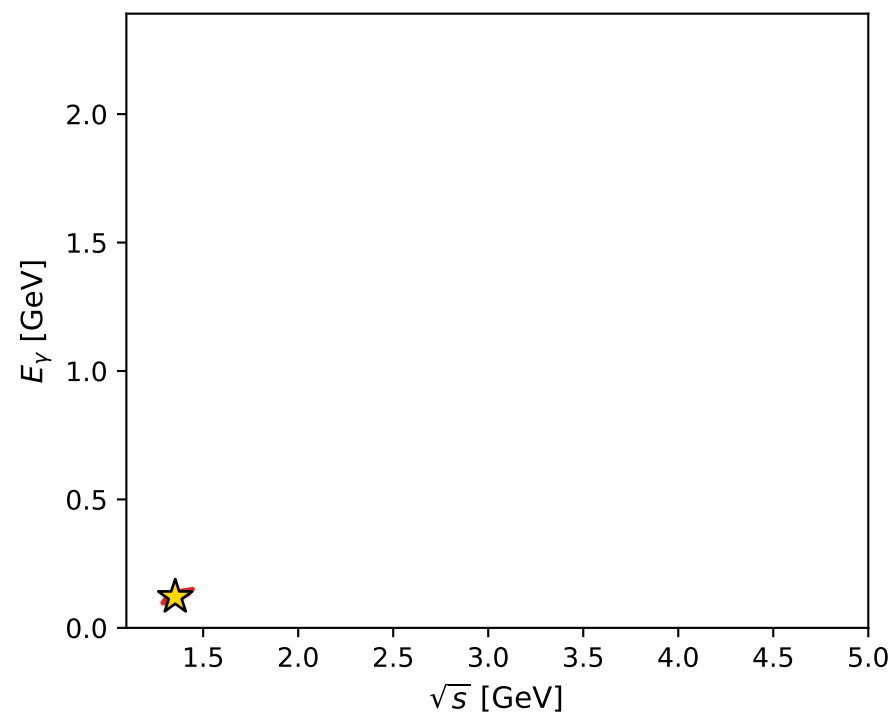
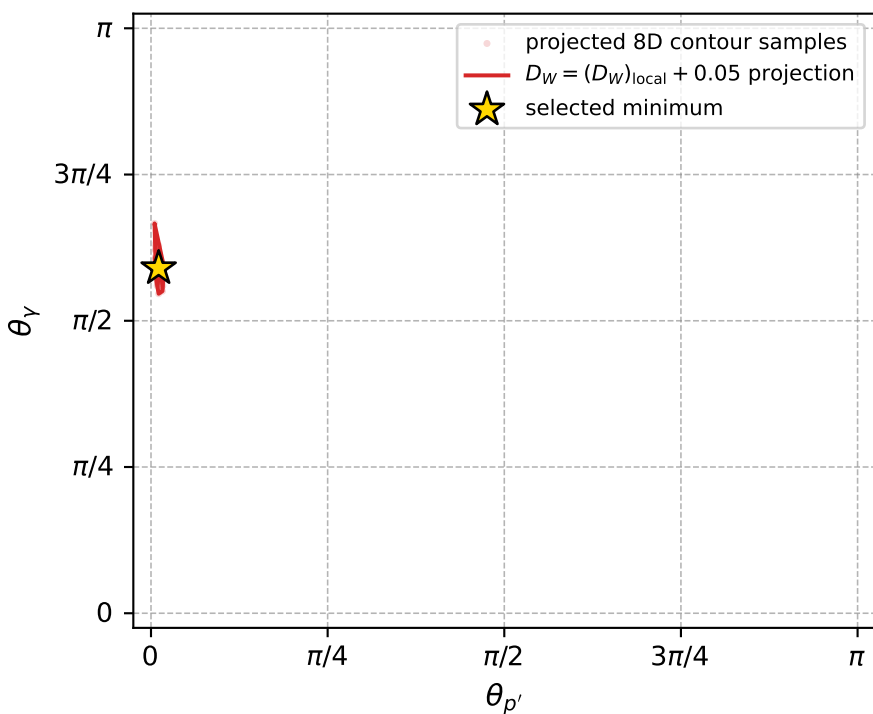
dw_mixing_angles_polarization_03_local_minimum_726 D_W=0.00373041
 region: polarization_cluster_03_local_minimum_726
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.80498483$ rad
 incoming lepton: $0.693122|+\rangle + 0.72082|-\rangle$
 proton mixing angle: $\alpha_p=2.7506222$ rad
 incoming proton: $-0.92454|+\rangle + 0.381086|-\rangle$

$s=1.8303$, $\sqrt{s}=1.35289$, $|\vec{P}|=0.339326$, $|\vec{P}'|=0.248274$
 $\theta_{P'}=0.0340283$, $\theta_V=1.85302$, $\phi_{P'}=1.5684$, $\phi_V=4.81623$
 $E_V=0.120438$, $\theta_{\ell'}=2.67801$, $\phi_{\ell'}=1.68299$
 $Q^2=0.0183726$, $x_B=0.0678747$, $t=-0.00764879$, $W^2=1.13216$, $y=0.288179$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.3554$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.3393$
 P $E= 0.9975$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.3393$
 ℓ' $E= 0.2621$ $p_x= -0.0120$ $p_y= 0.1066$ $p_z= -0.2146$
 P' $E= 0.9703$ $p_x= 0.0000$ $p_y= 0.0084$ $p_z= 0.2481$
 q_Y $E= 0.1204$ $p_x= 0.0120$ $p_y= -0.1151$ $p_z= -0.0335$



muon: polarization cluster P3, local minimum 726; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 0.0037304117$
 $D_W \text{ contour} = 0.053730412$
 above global minimum = 0.0037268789
 8D contour samples = 6561

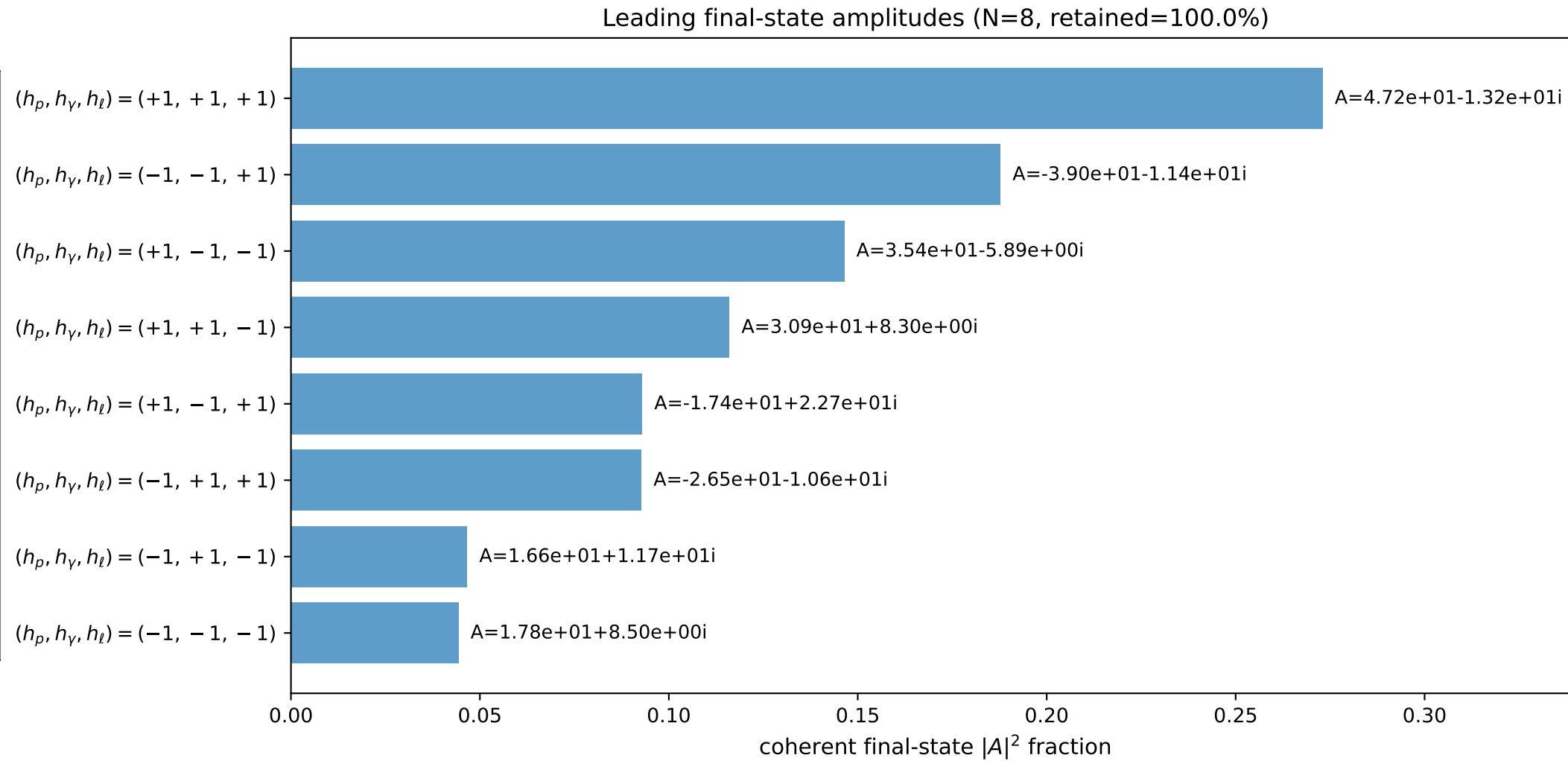
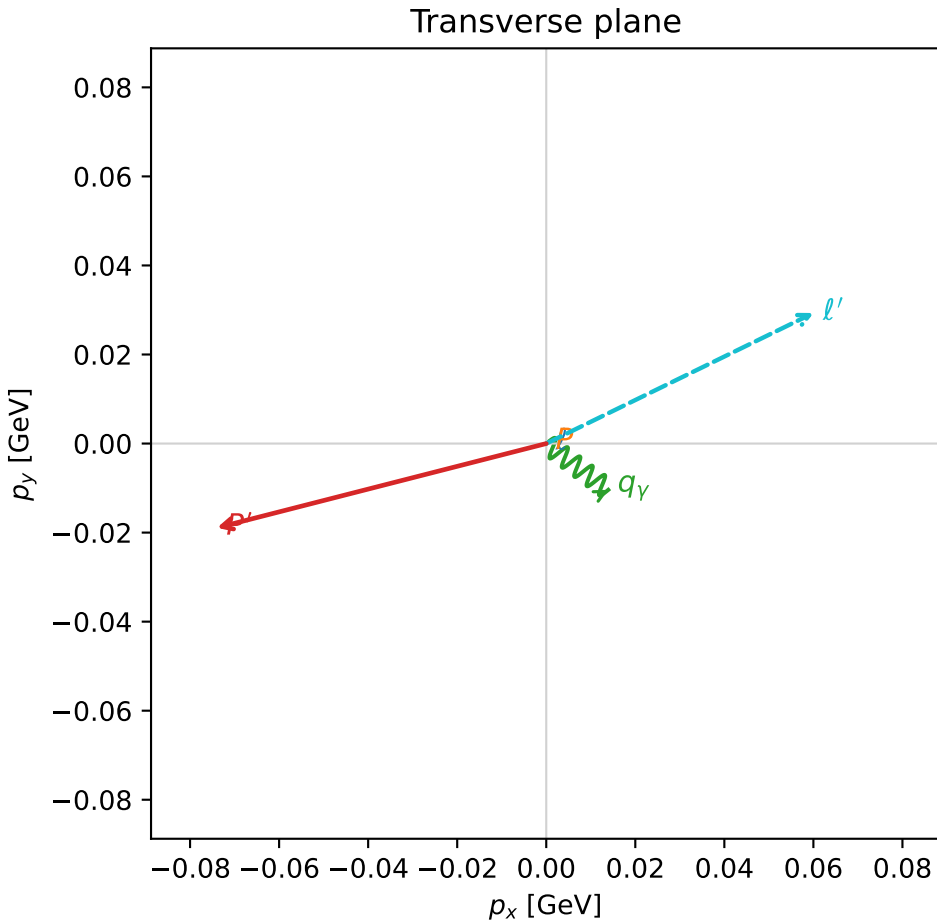
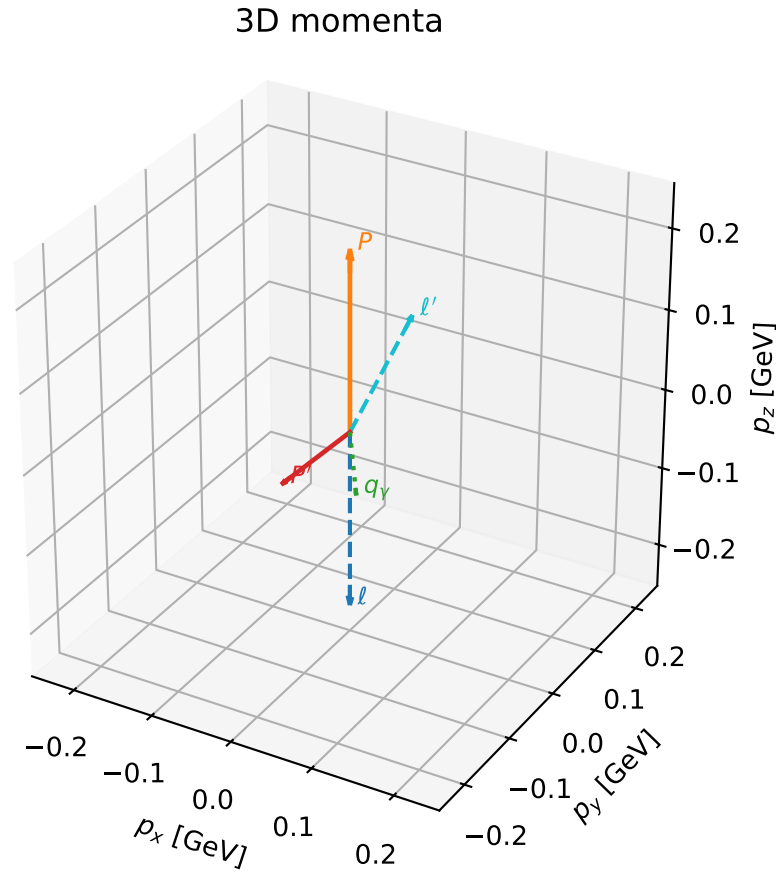
selected phase-space point:
 $\text{sqrt_s} = 1.352886$
 $\text{theta_p_out} = 0.03402827$
 $\text{theta_gamma_out} = 1.853025$
 $\text{qOut} = 0.1204384$
 $\text{phi_p_out} = 1.568398$
 $\text{phi_gamma_out} = 4.816229$
 $\text{alpha_e} = 0.8049848$
 $\text{alpha_p} = 2.750622$

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

dw_mixing_angles_polarization_03_local_minimum_734 D_W=0.00397504
 region: polarization_cluster_03_local_minimum_734
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=0.88639631$ rad
 incoming lepton: $0.632208|+\rangle + 0.774798|-\rangle$
 proton mixing angle: $\alpha_p=2.8869429$ rad
 incoming proton: $-0.967752|+\rangle + 0.251906|-\rangle$

$s=1.46169$, $\sqrt{s}=1.209$, $|\vec{P}|=0.22132$, $|\vec{P}'|=0.10806$
 $\theta_{P'}=2.35696$, $\theta_V=2.89835$, $\phi_{P'}=3.39203$, $\phi_V=5.64193$
 $E_V=0.0721878$, $\theta_{\ell'}=0.427815$, $\phi_{\ell'}=0.454336$
 $Q^2=0.137008$, $x_B=0.518419$, $t=-0.0941252$, $W^2=1.00712$, $y=0.463096$

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 0.2452$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= -0.2213$
 P $E= 0.9638$ $p_x= 0.0000$ $p_y= 0.0000$ $p_z= 0.2213$
 ℓ' $E= 0.1926$ $p_x= 0.0600$ $p_y= 0.0293$ $p_z= 0.1465$
 P' $E= 0.9442$ $p_x= -0.0740$ $p_y= -0.0189$ $p_z= -0.0765$
 q_Y $E= 0.0722$ $p_x= 0.0139$ $p_y= -0.0104$ $p_z= -0.0701$



muon: polarization cluster P3, local minimum 734; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour

